

BLE Bootloader & DFU DEMO for nRF54l15

Introduction

Demo use SEGGER Embedded Studio Release 8.24 Build 2025041401.59002

nrfx 3.11 , nRF54L15 DK.

BLE_APP_54l15

BLE_APP_54l15 is a simple BLE app example with SoftDevice S115.

https://github.com/guangli2015/BLE_APP_nRF54L15.git

Memory setup is in Setup\nrf54l15_xxaa_application.ld

MEMORY

```
{  
    FLASH (rx) : ORIGIN = 0x0000b000, LENGTH = 0x146800  
    RAM (rwx) : ORIGIN = 0x20004400, LENGTH = 0x3B800  
}
```

bootloader_54l15

bootloader_54l15 is bootloader DEMO in 54l15 which perform Jumping to APP or entering DFU mode to upgrade firmwareLoader

https://github.com/guangli2015/bootloader_54l15.git

Memory setup is in Setup\nrf54l15_xxaa_application.ld

MEMORY

```
{  
    FLASH (rx) : ORIGIN = 0x00000000, LENGTH = 0x9000 /* Inside global RRAM0 */  
    RAM (rwx) : ORIGIN = 0x20004400, LENGTH = 0x3B800
```

```
}
```

firmwareloader_54l15

firmwareloader_54l15 is firmwareloader DEMO in 54l15 which perform upgrading BLE app.

https://github.com/guangli2015/firmwareloader_54l15.git

MEMORY

```
{
```

```
    FLASH (rx) : ORIGIN = 0x00151800, LENGTH = 0x00010000 /* Inside global RRAM0 */
```

```
    RAM (rwx) : ORIGIN = 0x20004400, LENGTH = 0x3B800
```

```
}
```

Test step

Reset + button 0 in nrf54l15DK entering bootloader dfu mode

Reset + button 1 entering ble app

Reset entering firmwareloader dfu mode

DEMO HEXs downloading into nRF54L15DK

1. Open BLE_APP_54l15.emProject and build entire solution
2. Open bootloader_54l15.emProject and build entire solution
3. Open firmwareLoader_54l15.emProject and build entire solution
4. nrfutil device recover
5. nrfutil device erase
6. cd C:\work\bootloader_54l15\Output\Debug\Exe
7. nrfutil device program --firmware bootloader_54l15.hex --options
chip_erase_mode=ERASE_NONE
8. cd C:\work\BLE_APP_54l15\Output\Debug\Exe
9. nrfutil device program --firmware BLE_APP_54l15.hex --options
chip_erase_mode=ERASE_NONE
10. cd C:\work\firmwareloader_54l15\Output\Debug\Exe
11. nrfutil device program --firmware firmwareLoader_54l15.hex --options
chip_erase_mode=ERASE_NONE

12. cd C:\work\bootloader_54l15\Source\softdevice\s115

13. nrfutil device program --firmware s115_9.0.0-3.prototype_nrf54l15_softdevice.hex --options chip_erase_mode=ERASE_NONE



J-Link RTT Viewer V8.18
Compiled: 14:45:05 Mar 5 2025
<https://www.segger.com>

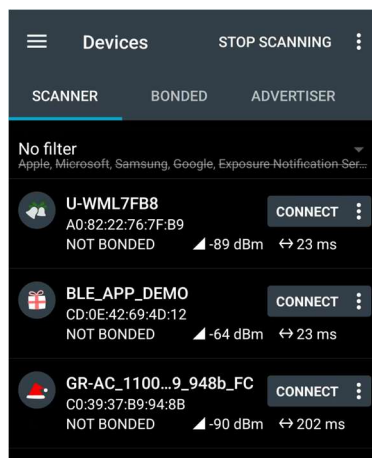
14. The debug log can be view using



15. Open (tested using nrf connect v4.29.1 android) in your phone

BLE app upgrade

16. Press Reset +button 1 in 54l15DK, Please check if have the adv device
BLE_APP_DEMO



17. Open BLE_APP_54l15.emProject again and change Source\prj_config.h

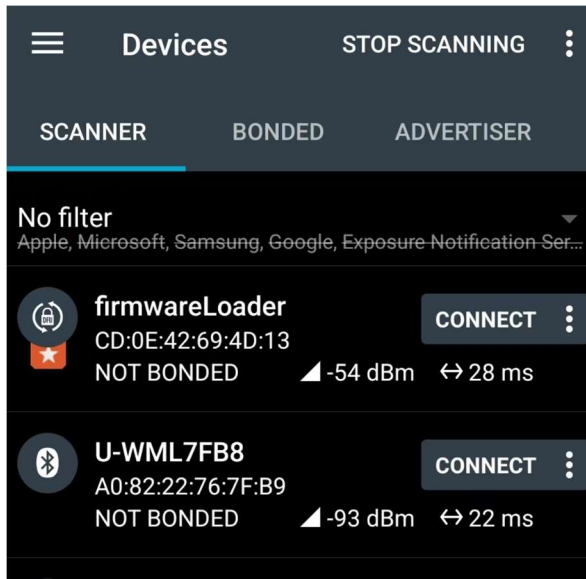
CONFIG_BLE_ADV_NAME "BLE_APP_DEMO" to #define CONFIG_BLE_ADV_NAME
"BLE_APP_DEMO_new" and build

18. Cd C:\work\BLE_APP_54l15\Output\Debug\Exe & change 'BLE_APP_54l15.hex' into
'BLE_APP_54l15_new.hex'

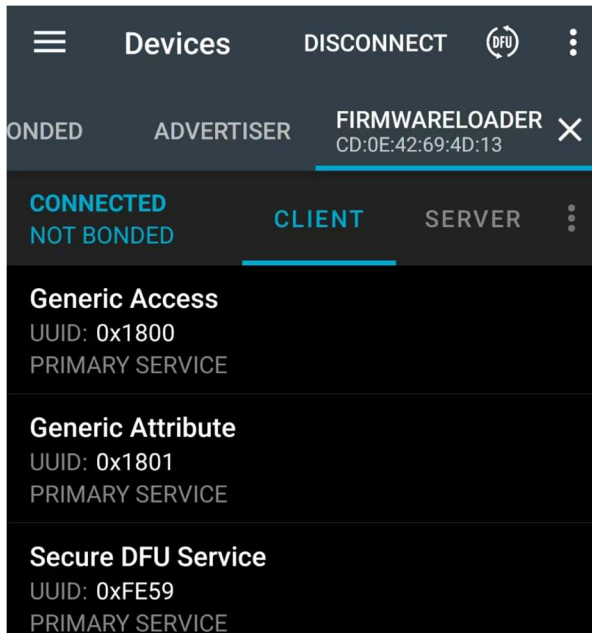
19. nrfutil nrf5sdk-tools keys generate priv.pem

20. `nrfutil nrf5sdk-tools pkg generate --application BLE_APP_54l15_new.hex --application-version 2 --hw-version 52 --sd-req 0x100 --key-file priv.pem BLE_APP_54l15_new.zip`
21. transfer 'BLE_APP_54l15_new.zip' into your phone
22. Press the reset in 54l15DK to enter firmwareloader dfu mode

Then you can see the 'firmwareLoader'



Connected and press DFU button in the up right corner

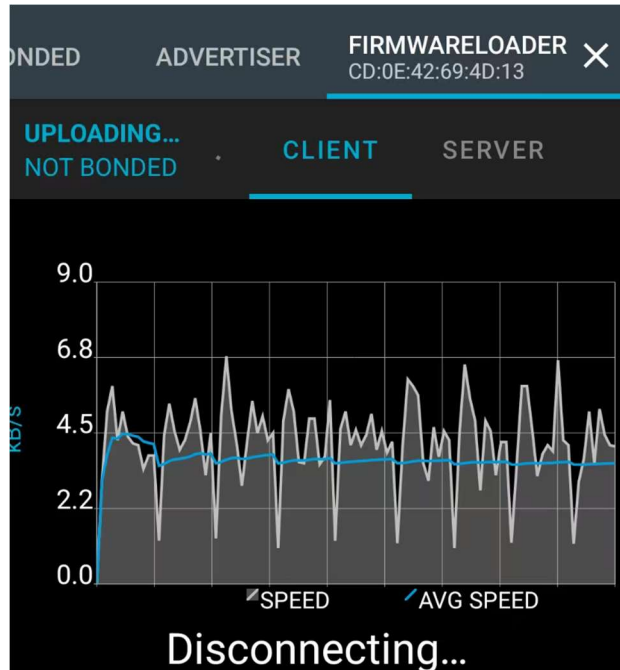


Select the 'BLE_APP_54l15_new.zip' in your phone



Distribution packet (ZIP)

Press Reset+ button 1 in your board manually after transfer finish



You will see the BLE_APP_DEMO_new

Devices

STOP SCANNING

SCANNER

BONDED

ADVERTISER

No filter

Apple, Microsoft, Samsung, Google, Exposure-Notification Ser...

BLE_APP_DEMO_new

CD:0E:42:69:4D:12

NOT BONDED

-60 dBm

↔ 24 ms

CONNECT

U-WML7FB8

A0:82:22:76:7F:B9

NOT BONDED

-85 dBm

↔ 23 ms

CONNECT

GR-AC_1100...9_948b_FC

C0:39:37:B9:94:8B

NOT BONDED

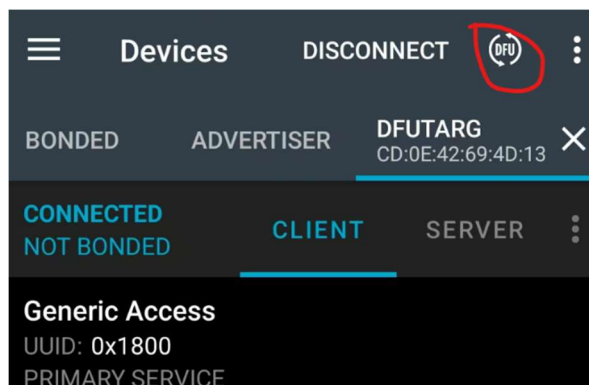
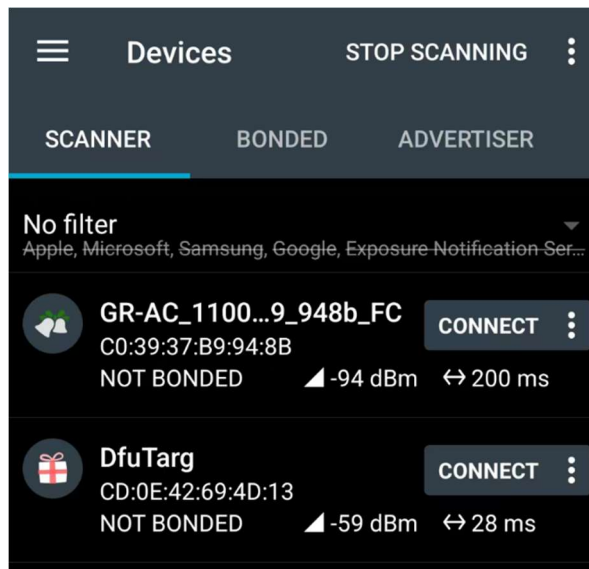
-91 dBm

↔ 210 ms

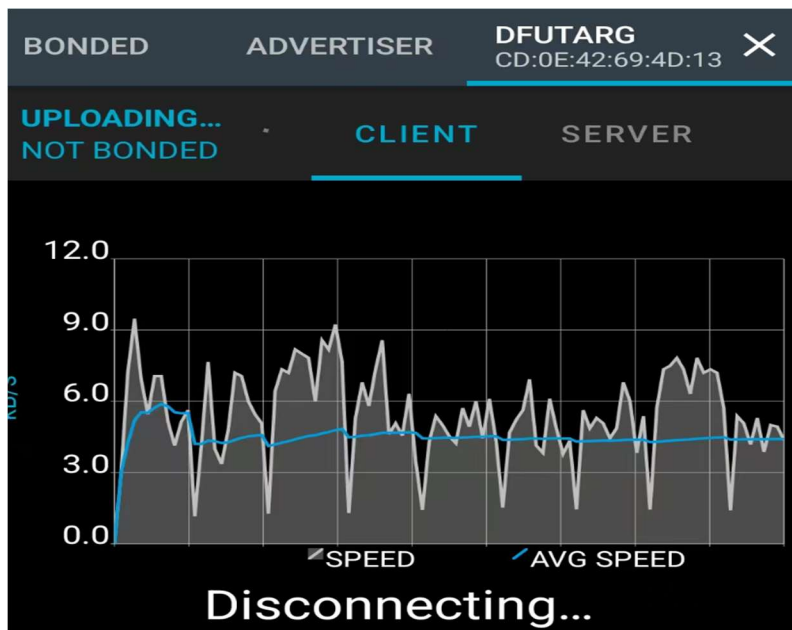
CONNECT

firmwareLoader upgrade

1. change firmwareloader_54l15\Source\ble_dfu\nrf_dfu_ble.c
#define NRF_DFU_BLE_ADV_NAME "firmwareLoader" into "firmwareLoader2027"
and build the firmwareloader_54l15 project
2. cd firmwareloader_54l15\Output\Debug\Exe & change
‘firmwareLoader_54l15.hex’ into ‘firmwareLoader_54l15_new.hex’
3. nrfutil nrf5sdk-tools keys generate priv.pem
4. nrfutil nrf5sdk-tools pkg generate --application firmwareLoader_54l15_new.hex -
-application-version 2 --hw-version 52 --sd-req 0x100 --key-file priv.pem
firmwareLoader_54l15_new.zip
5. transfer ‘firmwareLoader_54l15_new.zip’ into your phone
6. Press the reset+ button 0 in 54l15DK to enter bootloader dfu mode
7. Then you can see the ‘dfuTarg’



Select the ‘firmwareLoader_54l15_new.zip’ in your phone



8. Press Reset in your board manually after transfer finish
You will see the "firmwareLoader2027"

